

Statistical Analysis Explanation on whether Anaesthesiology Trainees Can Distinguish Between Benign and Pathologic Pre-Recorded Paediatric Cardiac Murmurs

Participant characteristics

Initial data processing consisted of a workflow which included the following:

- (i) harmonised column names,
- (ii) merged the “Results” and “Factors” worksheets on the quiz_group identifier,
- (iii) recoded experience-related fields into ordered factors,
- (iv) creating a new variable, performance, defined as *Adequate* for scores $\geq 60\%$ vs *inadequate* for scores $< 60\%$, and
- (v) removed records with incomplete key variables.

After cleaning, 35 anaesthesiology trainees remained for analysis (Table 1).

Most participants had substantial post-community-service experience, with 24/35 (69 %; 95 % Confidence Interval (CI) 51 – 83 %) reporting more than four years in practice. Conversely, only 11 (31 %; 95% CI: 17 – 49 %) had four or fewer years of experience.

Regarding progression through specialist training, 22 respondents (63 %; 95% CI: 45 – 78 %) had completed their anaesthesiology registrar programme within the preceding two years, whereas 13 (37 %; 22 – 55 %) had done so earlier.

Exposure to respective rotations varied:

- ICU rotation (≥ 3 months): 15/35 (43 %; 27 – 60 %)
- Cardiothoracic-anaesthesia rotation (≥ 3 months): 10/35 (29 %; 15 – 47 %)

General practice experience was common with 21 trainees (60 %; 42 – 76 %) who had spent more than two years as a GP. In contrast, extended experience in internal medicine/cardiology or paediatrics was rare, with only 1 (2.9 %; 0.2 – 17 %) and 2 (5.7 %; 1.0 – 21 %) individuals, respectively, exceeding two years in those disciplines.

Finally, when performance on the 20-item quiz was dichotomised, 20 trainees (57 %; 40 – 73 %) achieved an *Adequate* score ($\geq 60\%$ correct), while 15 (43 %; 27 – 60 %) fell below this threshold.

Table 1. Participant characteristics

Variable	N = 35 [†]	95% CI
Years experience post community service		

Variable	N = 35[†]	95% CI
>4 years	24 (69%)	51%, 83%
≤4 years	11 (31%)	17%, 49%
Years completed of anaesthesiology registrar		
>2 years	13 (37%)	22%, 55%
≤2 years	22 (63%)	45%, 78%
ICU rotations completed (3 months)	15 (43%)	27%, 60%
Cardiothoracic rotations completed (3 months)	10 (29%)	15%, 47%
Experience as a general practitioner		
>2years	21 (60%)	42%, 76%
≤2 years	14 (40%)	24%, 58%
Experience in internal medicine or cardiology		
>2 years	1 (2.9%)	0.15%, 17%
≤2 years	34 (97%)	83%, 100%

Variable	N = 35 ¹	95% CI
Experience in paediatrics		
>2 years	2 (5.7%)	1.0%, 21%
≤2 years	33 (94%)	79%, 99%
Performance		
Adequate	20 (57%)	40%, 73%
Inadequate	15 (43%)	27%, 60%
Abbreviation: CI = Confidence Interval		
¹ n (%)		

Comparison of Adequate vs Inadequate Performers

Table 2 contrasts trainees who achieved adequate diagnostic performance on the quiz ($\geq 60\%$ correct; $n = 20$) with those whose performance was inadequate ($< 60\%$ correct; $n = 15$).

Specificity and sensitivity differed significantly between groups. Specificity was higher among adequate performers (Mean = 0.57, SD = 0.15) than inadequate performers (Mean = 0.39, SD = 0.16; $p = 0.003$, Wilcoxon rank-sum), indicating greater ability to correctly identify benign murmurs. Similarly, mean sensitivity was greater in the adequate group (Mean = 0.75, SD = 0.14) compared with the non-adequate group (Mean = 0.62, SD = 0.14; $p = 0.007$).

In contrast, clinical experience variables showed no statistically significant association with diagnostic adequacy:

- Years of post-community-service practice
A majority in both groups reported > 4 years' experience (70 % vs 67 %; $p > 0.9$).

- Time completed registrar training
Distributions (> 2 years vs ≤ 2 years) were similar (40 % vs 33 %; $p = 0.7$).
- Rotational exposure
Proportions completing ≥ 3 months in ICU (45 % vs 40 %; $p = 0.8$) or cardiothoracic anaesthesia (30 % vs 27 %; $p > 0.9$) did not differ meaningfully.
- Prior general-practice experience
Identical split of > 2 years versus ≤ 2 years in both groups (60 % vs 40 %; $p > 0.9$).
- There was no experience in internal medicine/cardiology or paediatrics greater than 2 years and not associated with test performance (all $p \geq 0.4$).

Table 2. Results for adequate vs non-Adequate Performers

Variable	Adequate N = 20 ¹	inadequate N = 15 ¹	p- value ²
Specificity	0.57 (0.15)	0.39 (0.16)	0.003
Sensitivity	0.75 (0.14)	0.62 (0.14)	0.007
Years experience post community service			>0.9
>4 years	14 (70%)	10 (67%)	
≤ 4 years	6 (30%)	5 (33%)	
Years completed anaesthesiology registrar			0.7
>2 years	8 (40%)	5 (33%)	
≤ 2 years	12 (60%)	10 (67%)	
ICU rotations completed in 3 months	9 (45%)	6 (40%)	0.8
Cardiothoracic rotations completed in 3 months	6 (30%)	4 (27%)	>0.9

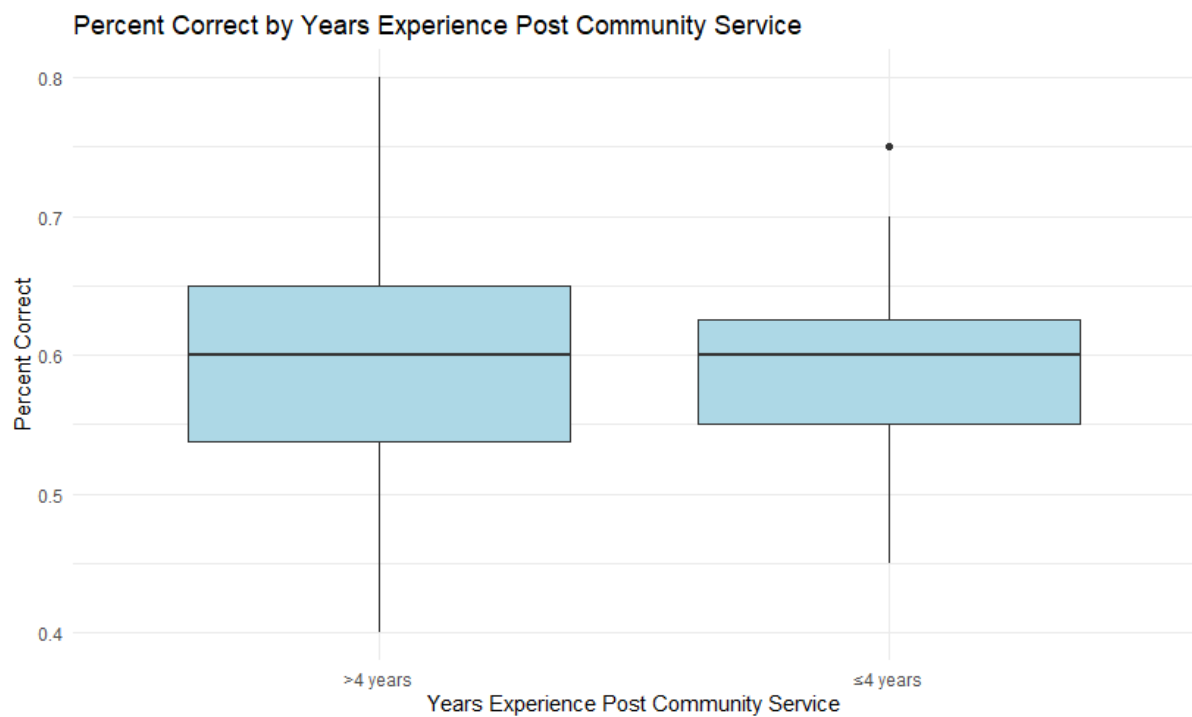
Variable	Adequate N = 20 ¹	inadequate N = 15 ¹	p- value ²
Experience as a general practitioner			>0.9
>2years	12 (60%)	9 (60%)	
≤2 years	8 (40%)	6 (40%)	
Experience in internal medicine or cardiology			0.4
>2 years	0 (0%)	1 (6.7%)	
≤2 years	20 (100%)	14 (93%)	
Experience in paediatrics			>0.9
>2 years	1 (5.0%)	1 (6.7%)	
≤2 years	19 (95%)	14 (93%)	
¹ Mean (SD); n (%)			
² Wilcoxon rank sum test; Fisher's exact test; Pearson's Chi-squared test			

Figures

Each of the following panels displays the distribution of quiz accuracy (percent correct) across two strata of a single exposure variable. The thick horizontal bar marks the median, the box denotes the inter-quartile range (IQR), and the whiskers extend to 1.5 × IQR (with outlying points shown individually). Hence, differences in medians and the overlap the IQRs provide a visual impression of between-group contrasts and within-group variability. We have excluded a plot related to the experience in internal medicine and cardiology as one of these categories had a zero response.

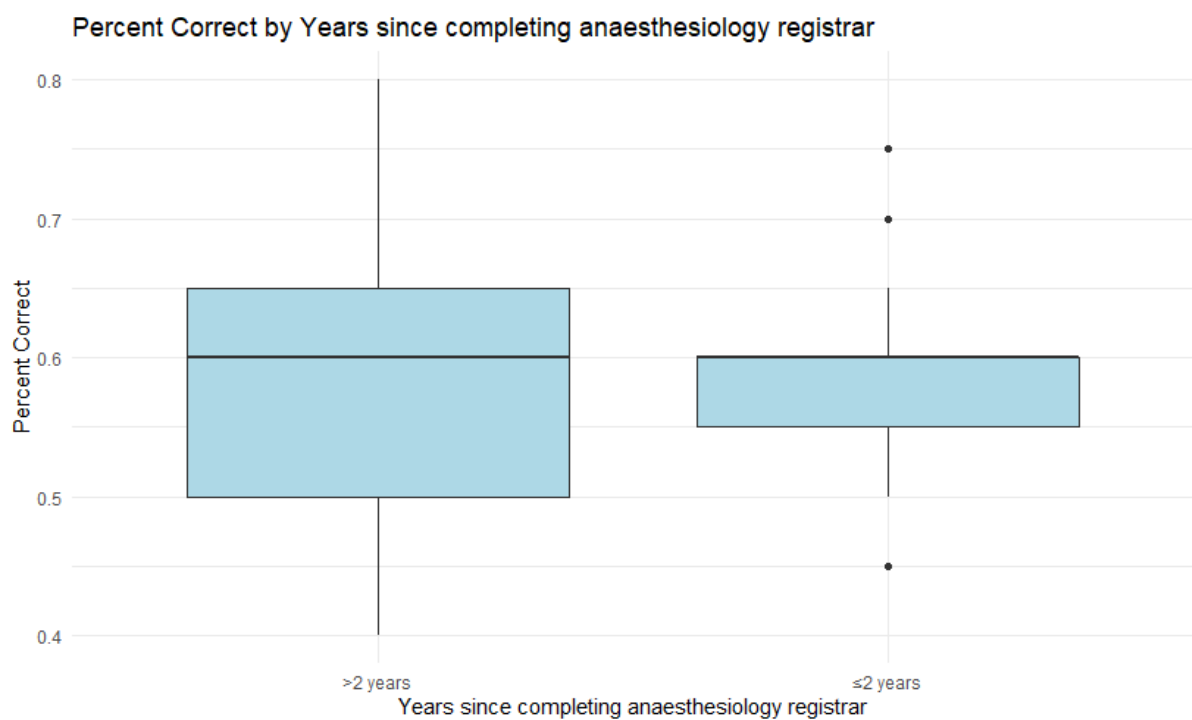
The median accuracy for number of year experience post-community service is roughly equal (~0.60) and spreads are also similar; extreme values in both groups reach ≥ 0.75.

Figure 1A.



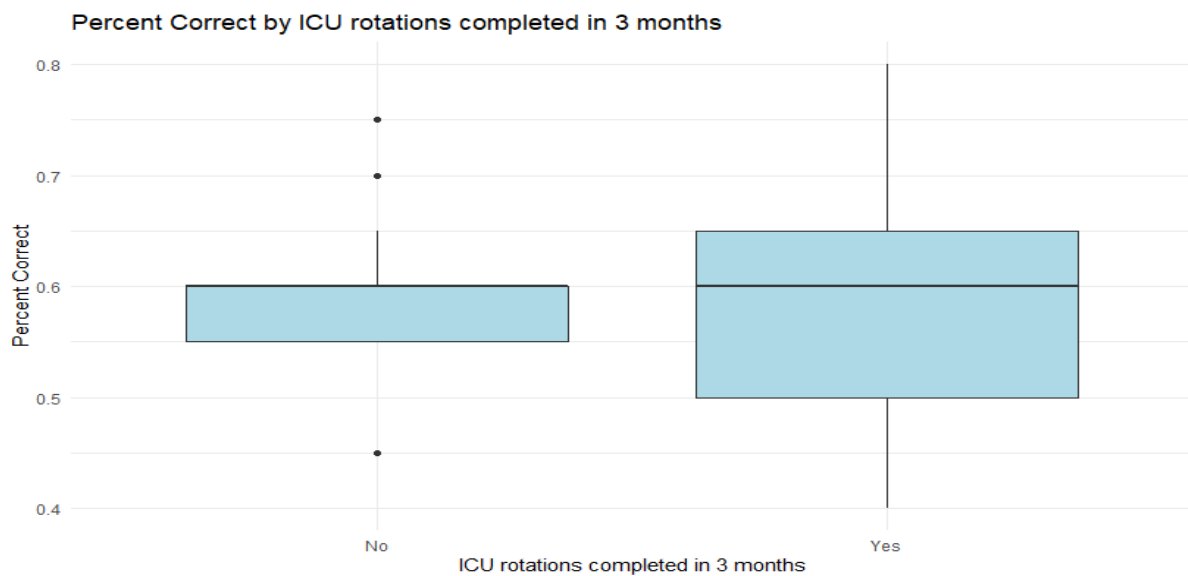
Median scores by years since completing their training as anaesthesiology registrars are identical (~0.60). There are slightly more outliers above 0.70 in the ≤ 2 -yr group, but overall IQRs overlap.

Figure 1B.



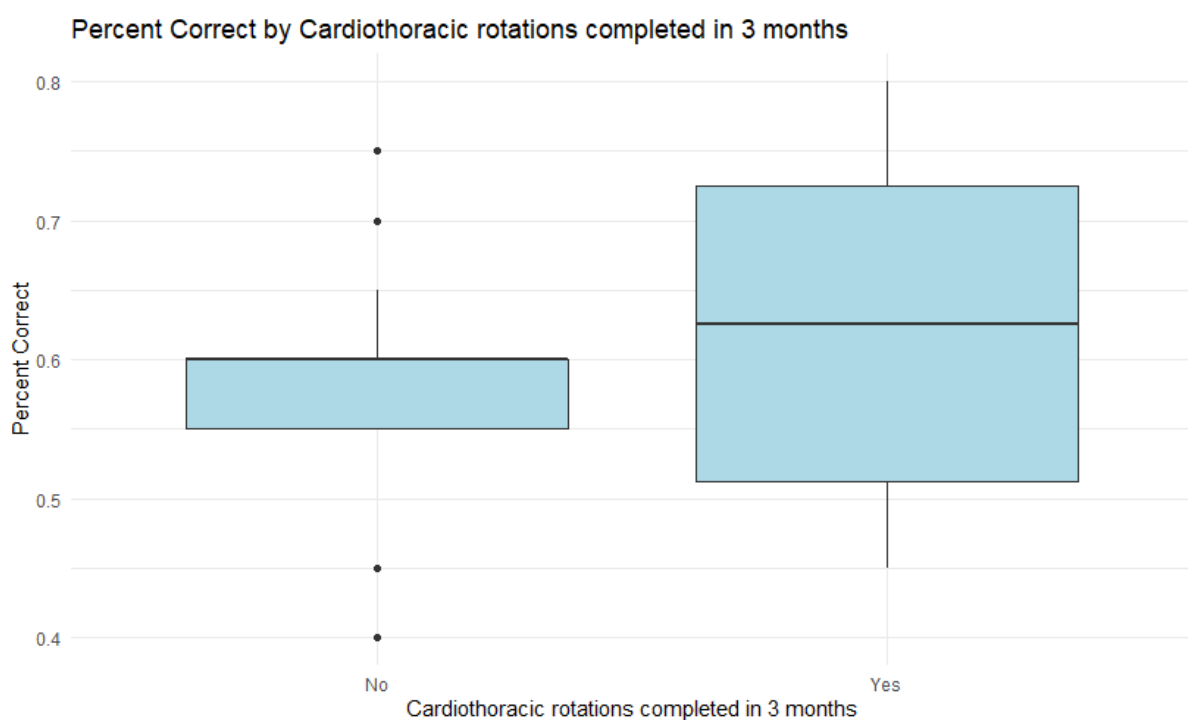
For those who had completed a rotation in the ICU within the last three months, median performance is the same (~0.60).

Figure 1C.



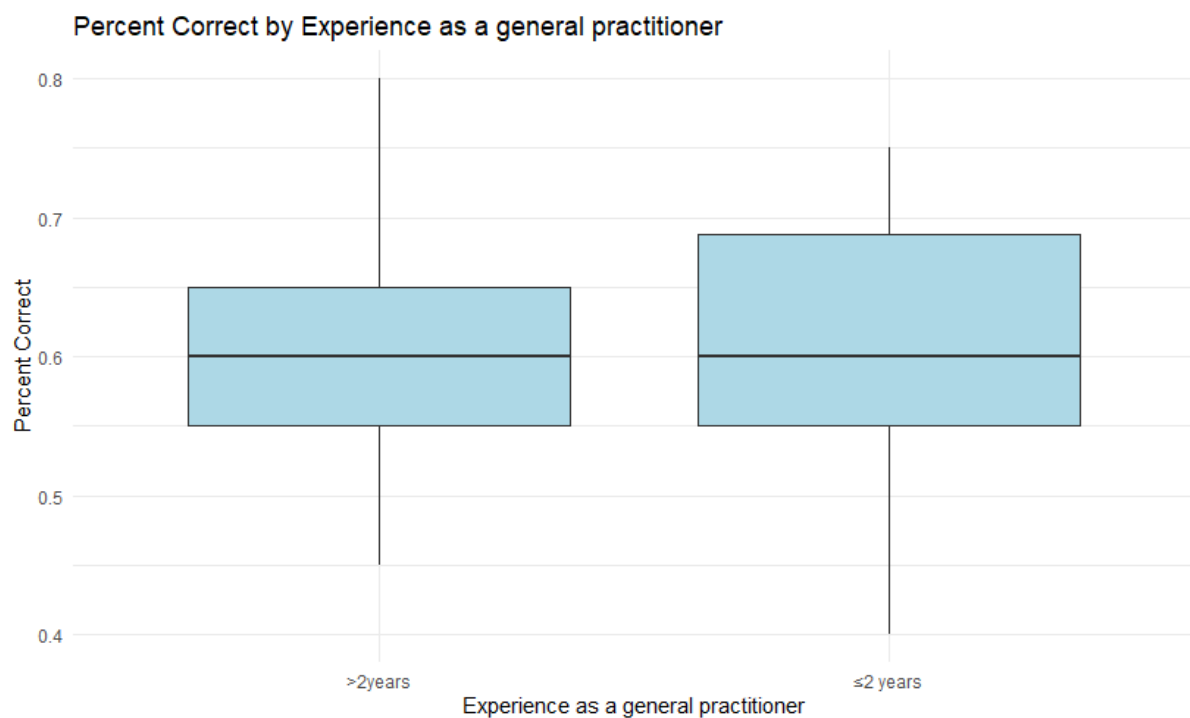
In terms of cardiothoracic rotation, the median accuracy is slightly higher for those with a completed rotation (~0.65 vs 0.60) and the upper quartile extends to ~0.75–0.80, but the IQRs overlap.

Figure 1D.



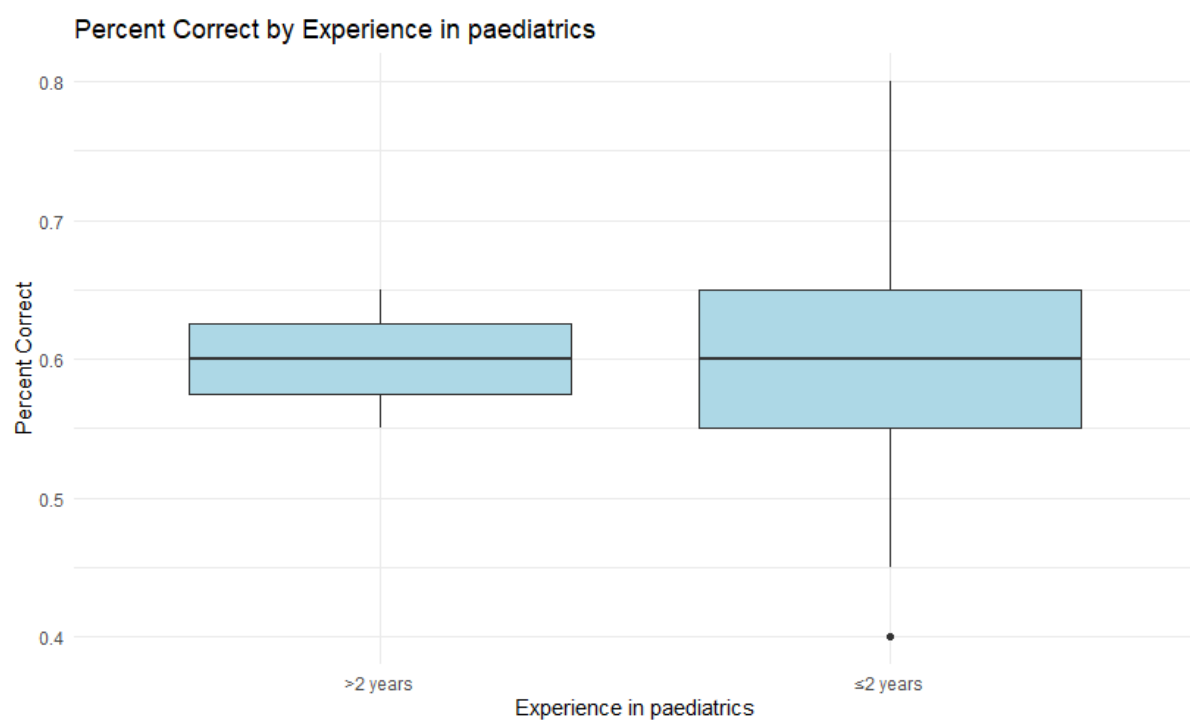
Both medians for percentage correct by experience as a general practitioner was ~0.60. The ≤ 2 -yr group shows slightly higher upper-quartile values, but again broad IQR overlap.

Figure 1E.



The median percentage scores sit at ~0.60 for both groups with respect to number of years experience in paediatrics, distributions overlap throughout.

Figure 1F



Univariable Predictors of Adequate Performance

Table 3 shows univariable logistic-regression results. Models were fitted to explore whether specific domains of clinical experience predicted an adequate score ($\geq 60\%$ correct). In each model the more experienced group served as the reference (OR = 1). Results are summarised below; all confidence intervals (CIs) are at the 95 % level.

- **Post-community-service experience.**
Trainees with ≤ 4 years of practice showed no meaningful difference in diagnostic success compared with those with > 4 years (OR = 1.17, 95 % CI 0.27–4.98; $p = 0.80$).
- **Time completed registrar training.**
Participants who had finished their registrar posts ≤ 2 years ago were not more likely to perform adequately than those > 2 years post-completion (OR = 1.33, 95 % CI 0.33–5.66; $p = 0.70$).
- **Critical-care rotations.**
Completion of a ≥ 3 -month ICU rotation (OR = 0.81, 95 % CI 0.20–3.17; $p = 0.80$) or cardiothoracic-anaesthesia rotation (OR = 0.85, 95 % CI 0.18–3.74; $p = 0.80$) was not associated with higher odds of adequate performance.
- **General-practice experience.**
Having ≤ 2 years versus > 2 years as a general practitioner yielded an OR of exactly 1.00 (95 % CI 0.25–3.95; $p > 0.90$), indicating the absence of a detectable relationship.
- **Internal-medicine/cardiology experience.**
No trainee in the sample reported > 2 years of internal-medicine exposure, producing a zero in the reference category. As a result, the model returned an OR of 0.00 for the ≤ 2 -year group and could not compute a CI. This results in us not being able to make a meaningful interpretation.
- **Paediatrics experience.**
Limited exposure to paediatrics (≤ 2 years) was associated with a non-significant reduction in the odds of adequate performance (OR = 0.74, 95 % CI 0.03–19.7; $p = 0.80$).

Across all domains, point estimates were near 1.0 and every 95 % CI included 1.0 within its upper and lower bounds, indicating no evidence that greater experience in particular fields of practice or rotation history influence quiz accuracy in this cohort.

Table 3. Univariate regression results

Variable	OR	95% CI	p-value
Post-community service experience			
>4 years	—	—	
≤4 years	1.17	0.27, 4.98	0.8
Time completed as anaesthesiology registrar			
>2 years	—	—	
≤2 years	1.33	0.33, 5.66	0.7
ICU rotation completed			
No	—	—	
Yes	0.81	0.20, 3.17	0.8
Cardiac rotation completed			
No	—	—	
Yes	0.85	0.18, 3.74	0.8
Experience as a general practitioner			
>2years	—	—	
≤2 years	1.00	0.25, 3.95	>0.9
Experience in internal medicine			
>2 years	—	—	

Variable	OR	95% CI	p-value
≤2 years	0.00		>0.9
Experience in paediatrics			
>2 years	—	—	
≤2 years	0.74	0.03, 19.7	0.8
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Multivariable Analysis of Factors Associated with Adequate Performance

A multivariable logistic-regression model was fitted to examine whether four domains of clinical exposure independently predicted an adequate quiz score. The variables *general-practice experience*, *internal-medicine/cardiology experience*, and *paediatric experience* were excluded because of sparsity in one or both categories or structural; including them produced unstable estimates. The remaining variables were included based on clinical relevance and evidence of association established in earlier analysis.

Table 4.

Variable	OR	95% CI	p-value
Post-community service experience			
>4 years	—	—	
≤4 years	1.04	0.17, 6.37	>0.9
Experience as an anaesthesiology registrar			
>2 years	—	—	
≤2 years	1.62	0.05, 49.1	0.8

Variable	OR	95% CI	p-value
ICU rotation completed			
No	—	—	
Yes	1.22	0.03, 44.7	>0.9
Cardiac rotation completed			
No	—	—	
Yes	1.05	0.11, 11.5	>0.9
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

None of the retained predictors showed a statistically significant association with diagnostic adequacy; all 95 % confidence intervals included the null value of 1 and *p*-values were well above the conventional 0.05 threshold. Point estimates hovered close to 1.0, suggesting little or no effect.